IRENA

International Renewable Energy Agency

RENEWABLE POWER GENERATION COSTS IN 2023

EXECUTIVE SUMMARY

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About IRENA

The International Renewable Energy Agency (IRENA) is an intergovernmental organisation that supports countries in their transition to a sustainable energy future and serves as the principal platform for international co-operation, a centre of excellence, and a repository of policy, technology, resource and financial knowledge on renewable energy. IRENA promotes the widespread adoption and sustainable use of all forms of renewable energy, including bioenergy, geothermal, hydropower, ocean, solar and wind energy, in the pursuit of sustainable development, energy access, energy security and low-carbon economic growth and prosperity. **www.irena.org**

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EXECUTIVE SUMMARY

HIGHLIGHTS

- Renewable power capacity additions set a record in 2023 with 473 GW of new installed capacity – a 54% increase compared to 2022 additions, and the largest annual growth since 2000.
- Total global renewables capacity in 2023 increased by 14% rate, from 3 391 GW in 2022 to 3 865 GW in 2023.
- In 2023, the global weighted average costs of electricity from newly-commissioned utility-scale solar photovoltaic (PV), onshore wind, offshore wind, concentrated solar power (CSP) and hydropower fell (Table S1).
- China represented the largest market for solar PV (63%), onshore wind (66%), offshore wind (65%) and hydropower (44%) in 2023. This was due to the country's substantial renewable additions in 2023, which drove the decline in the global weighted average costs for these technologies.
- In 2023, the total renewable power deployed globally since 2000 had saved an estimated USD 409 billion in fuel costs in the power sector.
- Battery storage annual capacity additions increased from 0.1 GWh gross capacity in 2010 to 95.9 GWh gross capacity in 2023. Between 2010 and 2023, the costs of battery storage projects declined 89%, from USD 2 511/kWh to USD 273/kWh.
- The competitiveness of renewable technologies remains, despite fossil fuel prices returning closer to their post-2010 cost range.
- In 2010, the global weighted average LCOE of onshore wind was 23% higher than the weighted average LCOE of fossil fuel; in 2023, the global weighted average LCOE of new onshore wind projects was 67% lower than the weighted average of those fossil fuel-fired solutions.
- In 2010, the global weighted average LCOE of solar PV was 414% higher than the weighted average LCOE of the cheapest fossil fuel-fired solution; however, driven by a spectacular decline in costs, in 2023, solar PV cost 56% less than the least-cost weighted average fossil fuel-fired solution.

Table S1 Total installed cost, capacity factor and LCOE trends by technology, 2010 and 2023

	Total installed costs (2023 USD/kW)			Capacity factor (%)			Levelised cost of electricity (2023 USD/kWh)		
	2010	2023	Percent change	2010	2023	Percent change	2010	2023	Percent change
Bioenergy	3 010	2 730	-9%	72	72	0%	0.084	0.072	-14%
Geothermal	3 011	4 589	52%	87	82	-6%	0.054	0.071	31%
Hydropower	1 459	2 806	92%	44	53	20%	0.043	0.057	33%
Solar PV	5 310	758	-86%	14	16	14%	0.460	0.044	-90%
CSP	10 453	6 589	-37%	30	55	83%	0.393	0.117	-70%
Onshore wind	2 272	1 160	-49%	27	36	33%	0.111	0.033	-70%
Offshore wind	5 409	2 800	-48%	38	41	8%	0.203	0.075	-63%

Notes: CSP = concentrated solar power; kW = kilowatt.

ANNUAL RENEWABLE POWER CAPACITY ADDITIONS BROKE A RECORD IN 2023, WITH TOTAL INSTALLED CAPACITY INCREASING 14%, YEAR-ON-YEAR.

In 2023, solar PV and onshore wind together represented more than 95% of the 473 GW in added renewable energy capacity.[1] Solar PV experienced an increase of 73% on 2023, adding 346 GW, while onshore wind added 104 GW, representing 48% year-on-year growth. Meanwhile, offshore wind capacity additions reached 11 GW, marking a 27% rise compared to 2022. This was, however, still below the record capacity additions of 2021 for this technology.

New additions were more modest for other technologies, such as concentrated solar power (CSP), geothermal, bioenergy and hydropower. Combined, these totalled 12 GW of additional installed capacity in 2023, of which 7 GW was hydropower. Annual additions for CSP and geothermal have been flat in recent years, while hydropower and bioenergy experienced a decrease in 2023 compared to 2022.

The growth in renewable power capacity additions reflects global efforts to transition the power sector to a higher share of renewables. New capacity additions, however, remain below the level needed to reach the goal of tripling capacity that was agreed in the UAE Consensus at COP28. Most importantly, the tripling of renewable power capacity must be accompanied by key enablers, notably grid expansion and storage.

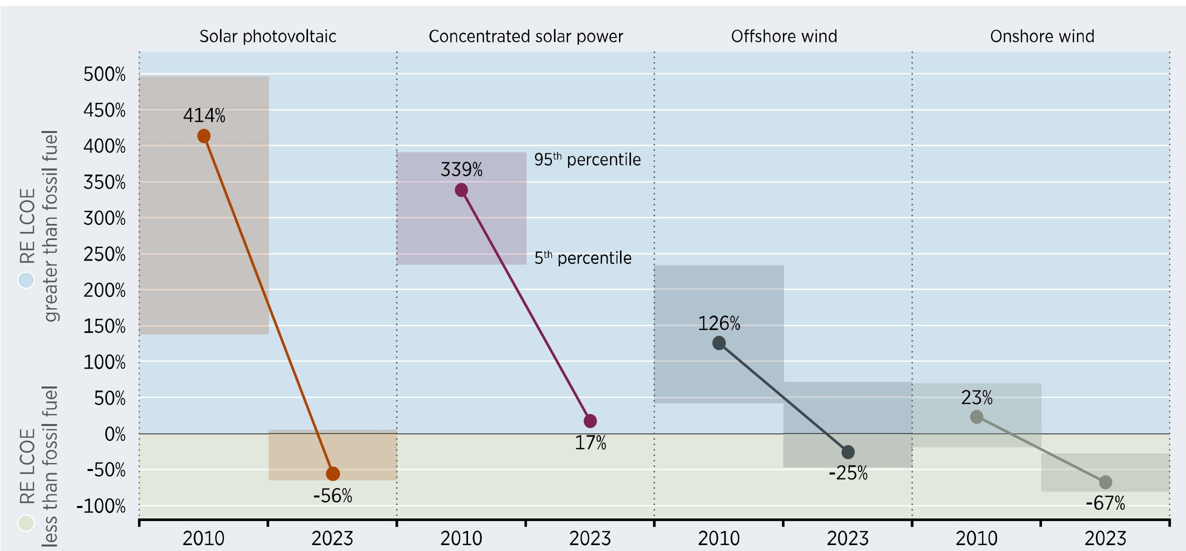
Data from the *IRENA Renewable Cost Database* and an analysis of recent power sector trends nonetheless reaffirms the essential role of renewables in reaching climate targets, while demonstrating the economic viability of these technologies compared to fossil fuels.

[1] *In this report, "renewable energy capacity" refers to the net generating capacity of power plants and other installations utilising renewable energy sources to produce electricity, commissioned within the respective year.*

After decades of falling costs and improving performance in solar and wind technologies, the economic benefits of renewable power generation – in addition to its social, developmental and environmental benefits – are now compelling.

In 2010, the global weighted average LCOE of onshore wind was USD 0.111/kWh. This was 23% higher than the weighted average cost of new capacity additions for fossil fuels*, which stood at USD 0.090/kWh. By 2023, however, the global weighted average LCOE of new onshore wind projects was USD 0.033/kWh – 67% lower than the weighted average fossil fuel-fired cost, which had risen to USD 0.100/kWh (Figure S2). Over the same period, the global weighted average LCOE of offshore wind went from being 126% more expensive than the weighted average fossil fuel cost to being 25% less expensive. The cost fell from USD 0.203/kWh to USD 0.075/kWh.

Figure S1 Change in global weighted average LCOE for solar and wind compared to fossil fuels, 2010-2023



Note: RE = renewable energy.

CSP, meanwhile, saw its global weighted average LCOE fall from being 339% higher than the weighted average fossil fuel option in 2010 to just 17% higher in 2023. Utility-scale solar PV had a global weighted average LCOE of USD 0.460/kWh in 2010 – 414% more expensive than the weighted average fossil fuel-fired option. Yet, by 2023, a spectacular decline in costs – to USD 0.044/kWh – left solar PV's global weighted average LCOE 56% lower than the weighted average fossil fuel-fired option.

Indeed, while 2023 saw fossil fuel-fired power generation costs fall from their high, 2022 values (Figure 1.6 and Figure 1.7), renewable power generation continued to be less expensive than fossil fuel options. In 2023, around 81% (382 GW) of newly-commissioned, utility-scale renewable power generation projects had costs of electricity lower than the weighted average fossil fuel-fired costs by country/region.

Overall, between 2010 and 2023, 1 690 GW of renewable power generation was deployed that had a lower LCOE than that of the weighted average fossil fuel-fired LCOE.

* *Note: In this report, the weighted average LCOE of renewables was compared to the weighted average fossil fuel LCOE, whereas in previous years this report has used the LCOE of the 'cheapest' fossil fuel-fired option. This is due to recent fossil fuel price reductions.*

RENEWABLE POWER BROUGHT BENEFITS TO THE ECONOMY.

Of the 20 countries[2] for which IRENA has detailed data, nine saw the competitiveness[3] of their utility-scale solar PV improve by more than the global weighted average LCOE in 2023. In 2022, eight countries saw such an improvement.

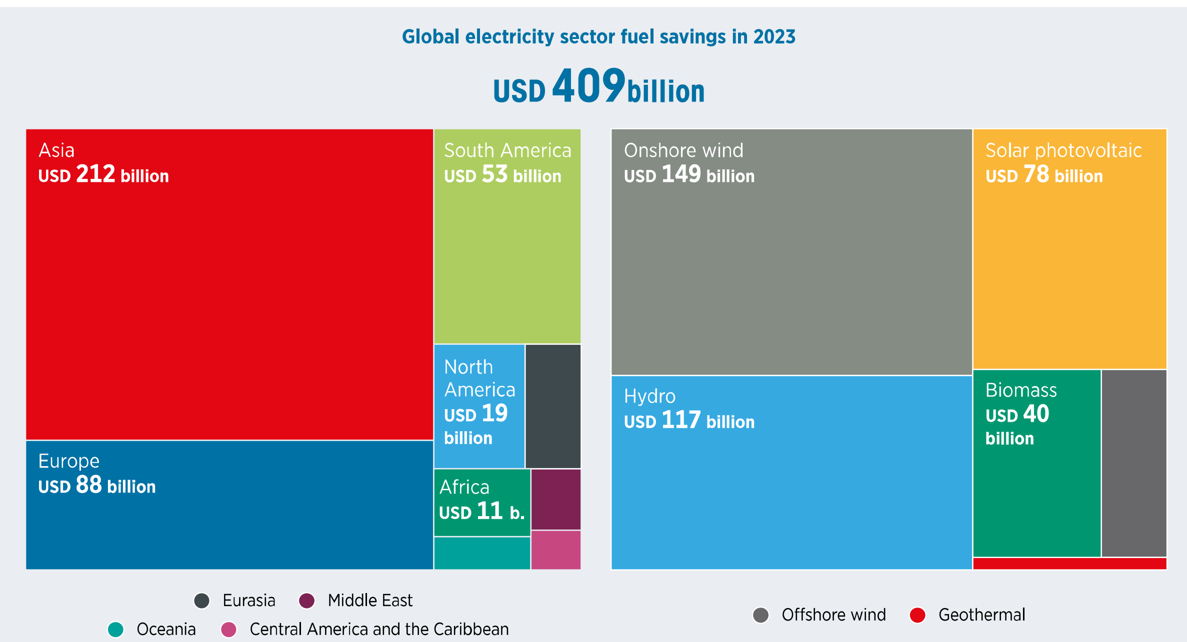
Over the 2022-2023 period, of the 19 countries examined for onshore wind, 16 saw the competitiveness of this technology improve by more than the global weighted average cost of electricity. In all markets, onshore wind was more competitive than the fossil fuel options.

Renewable power continues to save on fuel costs in the electricity sector. The competitiveness rate of solar and wind power was positive, although lower than in 2022. This was a consequence of the decline in fossil fuel prices.

In 2023, the renewable power deployed globally since 2000 saved an estimated USD 409 billion in fuel costs in the electricity sector alone (Figure S3). In the period between 2000 and 2010, Asia registered the highest cumulative savings, estimated at USD 212 billion. In Europe, the figure was USD 88 billion, followed by South America, where savings were estimated at USD 53 billion.

Regarding the technologies, onshore wind represented the highest savings, at USD 149 billion. Hydropower saw the second highest savings, at USD 117 billion, followed by solar PV, with USD 78 billion.

Figure S2 Global fossil fuel cost savings in the electricity sector in 2023 from renewable power added since 2000



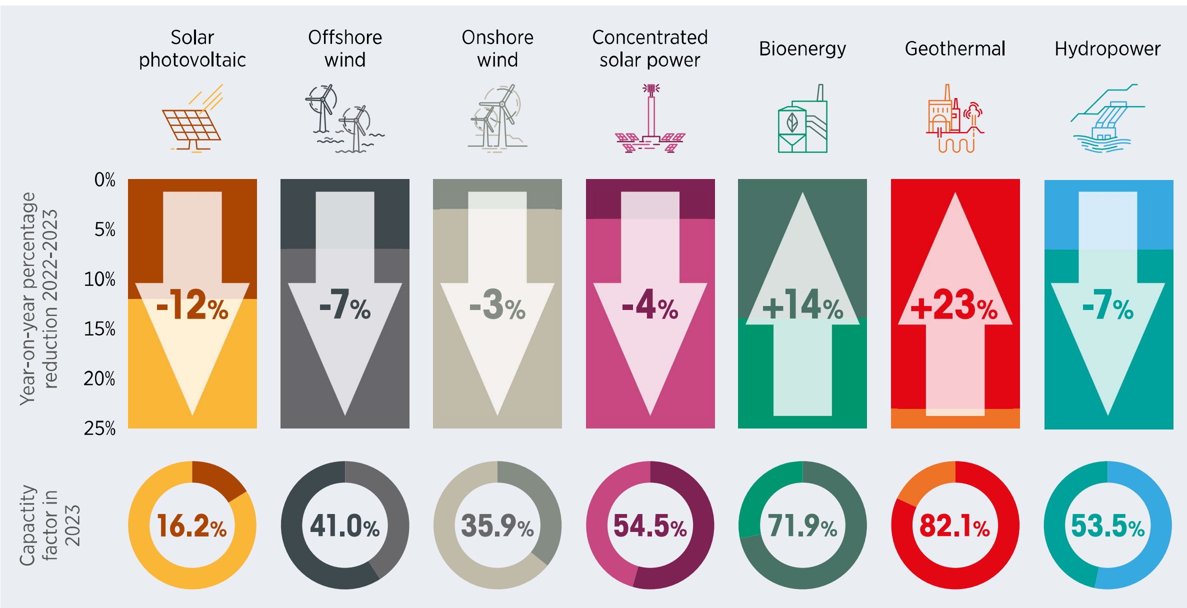
[2] *Detailed data is presented in Figure 1.10 covering the following 20 countries: Argentina, Australia, Brazil, Canada, China, France, Germany, India, Indonesia, Italy, Japan, Republic of Korea, Malaysia, Mexico, Philippines, South Africa, Türkiye, the United Kingdom, the United States and Viet Nam.*

[3] *IRENA has calculated a competitiveness metric for 20 countries. This is based on a weighted average cost of new fossil fuels calculated from project-level capital cost data and country-specific fossil gas and coal fuel marker prices to electricity generators. The competitiveness metric subtracts the country weighted average fossil fuel LCOE from the renewable power LCOE, so negative values represent renewable power LCOEs lower than those of fossil fuels.*

Between 2022 and 2023, the global weighted average total installed cost of newly-commissioned onshore wind projects decreased 13%, from USD 1 322/kilowatt (kW) to USD 1 154/kW. Over the same period, the global weighted average LCOE for these projects fell by 3%, from USD 0.035/kWh to USD 0.033/kWh (Figure S4).

In 2023, China was once again the largest market for new onshore wind capacity additions, with its share of global new deployment rising from 50% to 66%, year-on-year. This resulted in markets with higher installed costs decreasing their share relative to 2021. If China had been excluded, the global weighted average LCOE curve for onshore wind for the period would have increased 15%.

Figure S3 Global weighted average LCOE reduction and capacity factor from newly commissioned, utility-scale renewable power technologies, 2023



Note: The colour shading indicates the year-on-year percentage LCOE reduction (increase or decrease), starting from top (0%) to bottom (25%).

For newly-commissioned, utility-scale solar PV projects, the global weighted average LCOE decreased by 12% between 2022 and 2023, to USD 0.044/kWh. This was driven by a 17% decline in the global weighted average total installed cost for this technology, from USD 908/kW in 2022 to USD 758/kW for the projects commissioned in 2023.

Overall, 2023 saw solar PV experience a total installed cost decline in major markets. This was due to supply chain easing and reductions in commodity price inflation. European countries registered the biggest decrease in installed costs, with Greece seeing a decline of 48%, the Netherlands 41% and Germany 29%. The biggest markets followed the same trend, including China, which saw a decline of 10%, the United States (4%) and Brazil (5%). India was the exception, recording a 7% increase during 2023.

The offshore wind market added 11 GW of new capacity in 2023 – the second-highest year on record since 2021. China accounted for 65% of the total offshore capacity additions. Indeed, driven by China's share in new capacity additions and the commissioning of projects in new markets, the global weighted average cost of electricity of new projects saw a decrease of 7% in 2023, compared to 2022, from USD 0.080/kWh to USD 0.075/kWh.

Between 2022 and 2023, the global weighted average total installed costs of offshore wind decreased from USD 3 478/kW to USD 2 800/kW, while the weighted average capacity factor for newly-commissioned projects fell slightly, from 42% in 2022 to 41% in 2023.

In 2023, only one CSP project was completed. This resulted in a global weighted average LCOE of USD 0.117/kWh for this technology, representing a 4% decrease compared to 2022.

For newly-commissioned bioenergy for power projects, the global weighted average LCOE rose by 14% between 2022 and 2023, from USD 0.063/kWh to USD 0.072/kWh.

For geothermal power projects, between 2022 and 2023 the global weighted average LCOE of the seven projects commissioned increased by 23%, to USD 0.071/kWh.

Newly-commissioned hydropower projects, in contrast, saw their global weighted average LCOE decrease by 7% between 2022 and 2023, from USD 0.061/kWh to USD 0.057/kWh. Over the same period, the global weighted average total installed cost of new hydropower projects decreased from USD 3 053/kW to USD 2 806/kW – a fall of 8%.

SUBSTANTIAL COST REDUCTIONS IN RENEWABLES FROM 2010 TO 2023 DEMONSTRATE A REMARKABLE RATE OF DEFLATION.

Since 2010, solar PV has experienced the most rapid cost reductions. The global weighted average LCOE of newly-commissioned utility-scale solar PV projects declined from USD 0.460/kWh to USD 0.044/kWh between 2010 and 2023 – a decrease of 90% (Figure S5). This reduction in LCOE has been primarily driven by declines in module prices. These fell by around 93% between December 2009 and December 2023. Between 2018 and 2023, important reductions also occurred in soft costs (59%), modules and inverters (46%), balance of system (BoS) hardware (39%), and installation costs (36%). The median for all-in operations and maintenance (O&M) costs for utility-scale solar PV also decreased 5% between 2022 and 2023.

For onshore wind projects, the global weighted average cost of electricity fell by 70% between 2010 and 2023, from USD 0.111/kWh to USD 0.033/kWh. Cost reductions for onshore wind were driven by two key factors: wind turbine cost declines and capacity factor increases from turbine technology improvements.

Between 2010 and 2023, wind turbine prices outside China fell by between 41% and 64%, depending on the wind turbine price index. Within China, the decline was 73% for the same period. Meanwhile, the global weighted average capacity factor of newly-commissioned onshore wind projects increased from 27% in 2010 to 36% in 2023. This highlighted how technological improvements and cost reductions have made turbine installations cost competitive, even in areas with less favourable wind resources.

For newly-commissioned offshore wind projects, between 2010 and 2023 the global weighted average LCOE declined from USD 0.203/kWh to USD 0.075/kWh, a reduction of 63%. In 2010, China and Europe saw newly-commissioned offshore projects with a weighted average LCOE of USD 0.196/kWh and USD 0.205/kWh, respectively. The weighted average LCOEs of these two groups thereafter diverged, notably in 2021, when newly-commissioned European projects had a weighted average cost of USD 0.057/kWh – lower than the USD 0.085/kWh recorded in China that year. In 2023, the weighted average LCOE in Europe increased to USD 0.066/kWh as a range of more expensive projects were completed, including some in new markets. Europe's LCOE was still around 6% lower than the Chinese projects completed in 2023, which saw a weighted average of USD 0.070/kWh. The difference lies in the wind resource of each region, with Europe having higher wind speeds compared to those in China.

CSP deployment remains stagnant, with only 300 MW added in 2023 and global cumulative capacity standing at 7 GW at the end of 2023. For the period 2010 to 2023, the global weighted average cost of newly-commissioned CSP projects fell from USD 0.39/kWh to USD 0.117/kWh – a decline of 70%. The LCOE of CSP fell rapidly between 2010 and 2020, despite annual volatility. Since 2020, however, the commissioning of projects that were either delayed or included novel designs has seen the global weighted average cost of electricity from this technology stagnate.

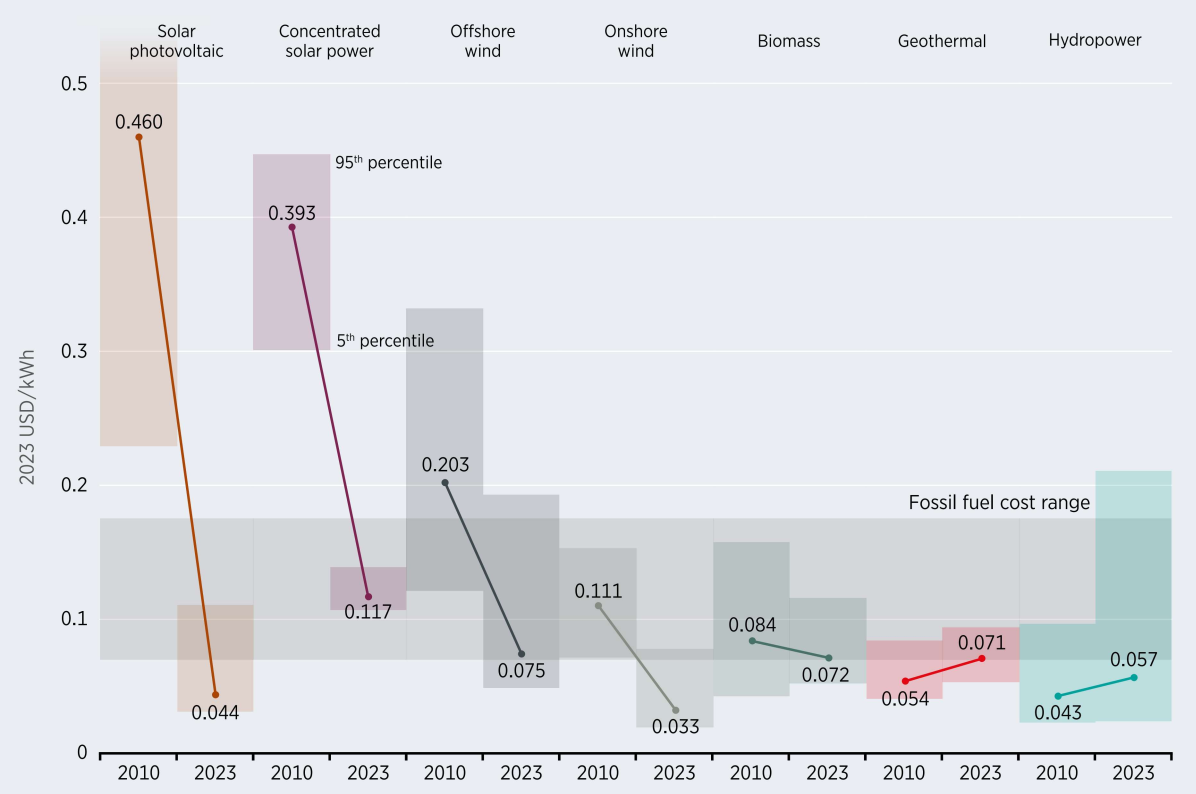
Bioenergy for power's global weighted average LCOE of USD 0.084/kWh in 2023 was 14% higher than the 2022 value and one-quarter lower than the USD 0.072/kWh value recorded in 2010. The increment is due to a shift in market share from 2022, with higher-cost markets now accounting for a greater share. Additionally, between 2010 and 2023 the global weighted average LCOE of bioenergy for power projects had experienced some volatility, without a notable trend upwards or downwards.

For geothermal projects, the global weighted average LCOE was 23% higher in 2023 than in 2022, reaching USD 0.071/kWh. This was still well within the USD 0.077/kWh to USD 0.074/kWh range seen between 2017 and 2021.

Newly-commissioned hydropower projects saw their global weighted average LCOE rise by 33% between 2010 and 2023, from USD 0.043/kWh to USD 0.057/kWh. This was still lower than the average fossil fuel-fired electricity option in 2023. During the period 2022 to 2023, the global weighted average costs decreased by 7%. The spike in costs in 2022 was driven by the commissioning of a number of projects that experienced very significant cost overruns, notably in Canada and the United States.

Electricity storage saw the costs of battery storage projects declined 89% between 2010 and 2023, from USD 2 511/kWh to USD 273/kWh. The cost reduction was driven by scaling up manufacturing, improved materials efficiency and improved manufacturing processes. Additionally, annual capacity additions increased from 0.1 GWh gross capacity in 2010 to 95.9 GWh gross capacity in 2023, with China accounting for almost half of the total global additions (46.5 GWh).

Figure S4 Global LCOE from newly-commissioned, utility-scale renewable power technologies, 2010 and 2023



Note: These data are for the year of commissioning. The thick lines are the global weighted average LCOE value derived from the individual plants commissioned in each year. The LCOE is calculated with project-specific installed costs and capacity factors, while the other assumptions, including weighted average cost of capital (WACC), are detailed in Annex I. The grey band represents the fossil fuel-fired power generation cost in 2023, while the bands for each technology and year represent the 5th and 95th percentile bands for renewable projects.

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